

SOME LIKE IT HOT: INCLUSIVE MONETARY POLICY UNDER OKUN'S HYPOTHESIS

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“The sacrifice of upward mobility must be carefully reckoned as one part of the high cost of accepting slack as a policy against inflation”

– Author M. Okun (BPEA 1973)

- Okun (1973): A high-pressure economy can **persistently** improve the economic circumstances of low-skill workers, allowing them to find steady employment and climb up the job ladder.
- Fed seems to increasingly adopt ‘inclusiveness’ into policy

FED'S NEW MONETARY POLICY FRAMEWORK

Fed seems to increasingly adopt 'inclusiveness' into policy Speech

- Maximum employment is a **broad-based** and **inclusive** goal
- "Our policy decision will be informed by **shortfalls** of employment rather than by **deviations**"
- Post-pandemic recovery: high inflation but also tight labor market even for low-skilled workers

Motivated by this policy shift, this paper

- Formalizes Okun's hypothesis by identifying 3 channels that give rise to Okun's hypothesis
- Builds a HANK Model + Frictional Labor Market (E,U,N) with these 3 channels to quantify inflation-inclusion trade-off
- Simulates counterfactual 'inclusive' monetary policy rules
 - Are Taylor rule or AIT inclusive? If not, what rules are?
 - Assess distributional and aggregate implications of the different MP

Key: Inflation vs. Inclusion tradeoff

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Key: Inflation vs. Inclusion tradeoff

PREVIEW OF THE RESULT

- AIT rule does **not** seem to be an inclusive rule
- An inclusive rule is an '**asymmetric** policy rule'

$$i_t = i^* + 1.25 (\pi_t - \pi^*) + 0.0 \log \left(\frac{Y_t}{Y^*} \right)^+ + 0.05 \log \left(\frac{Y_t}{Y^*} \right)^-$$

- At the cost of 1 ppt of additional inflation, under this rule:
 - Average unemployment declines by 0.6 ppt, labor force participation increases by 0.5 ppt
 - Larger positive effects for the bottom of the income distribution
 - At P25, unemployment declines by 1.3 ppt and 2 ppts increase in participation
 - Earnings inequality (P90 over P10) reduces by over 6 ppts

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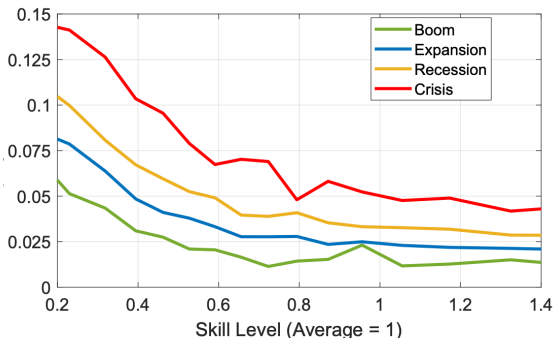
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Three Mechanisms of Okun's Hypothesis: Evidence

THREE CHANNELS OF OKUN'S HYPOTHESIS

- 1. Uneven exposures to business cycles (Aaronson et al., 2019)



- Low-skill workers more likely to be displaced in recession.
- High-pressure economy is especially beneficial to low-income groups

THREE CHANNELS OF OKUN'S HYPOTHESIS

- 2. Human capital accumulates over-time (Davis-von Wachter, 2011)
 - Stable employment → earnings growth
 - Unemployment → earnings decline (Persistent scarring of recession)
 - High-pressure economy raises (or limits the loss of) human capital stock.

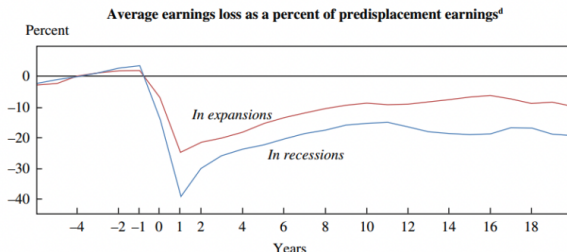
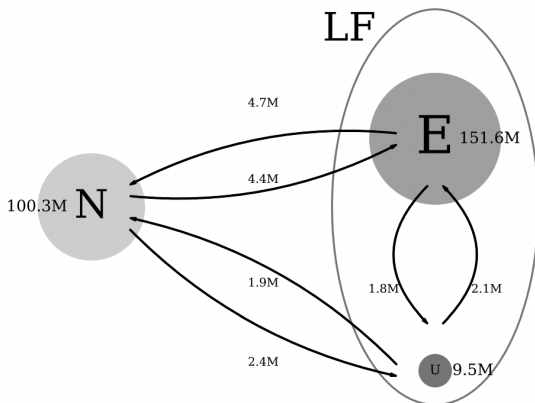


FIGURE: Average Earnings Loss upon Displacement

THREE CHANNELS OF OKUN'S HYPOTHESIS

- 3. Cyclical Participation Cycle (Hobijn-Sahin, 2021)
 - During recession, labor force participation declines
 - Increase in unemployment drives the decline ($UN \gg EN$)
 - Workers more likely to lose attachment while unemployed



THREE CHANNELS IN A NUTSHELL

In recessions,

- 1. A low-skilled worker more likely fired (Channel 1: Uneven exposure)
- 2. By staying unemployed, more likely to lose attachment to labor force and stop searching (Channel 3: Attachment)
- 3. Skills depreciate over time and employment becomes harder and less attractive (Channel 2: Human capital dynamics)
- Using HANK model with (E,U,N) and these three features, show hot economy disproportionately benefits low-skilled workers

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Model

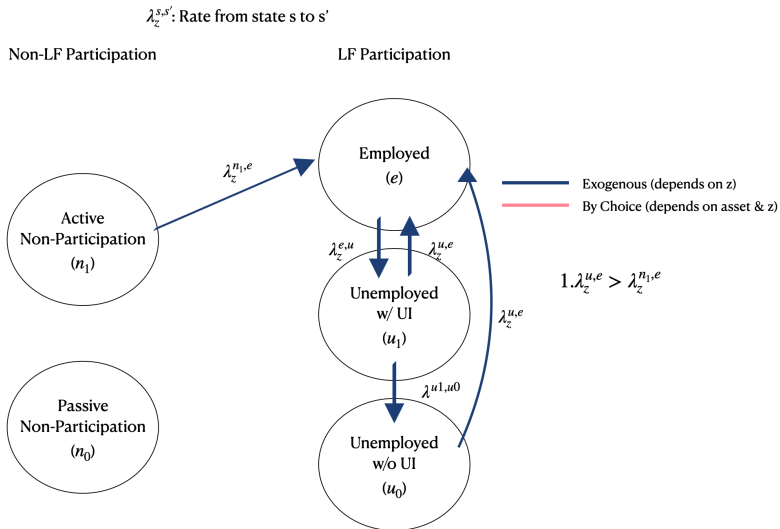
MODEL: INDIVIDUAL WORKER

- Skill level: z
- Labor market state: s

$$s = \begin{cases} e, & \text{employed} \\ u_0, & \text{unemployed, ineligible for UI} \\ u_1, & \text{unemployed, eligible for UI} \end{cases} \quad \begin{matrix} n_0, & \text{passive non-participant} \\ n_1, & \text{active non-participant} \end{matrix}$$

- Transition across labor market states:
 - **Exogenous** $e \rightarrow u, u \rightarrow e, n_1 \rightarrow e$ as a **function of skills z** (Channel 1: uneven exposure) Flow
 - Exogenous switch into and out of passive non-participation n_0
 - **Endogenous** participation choices: $n_1 \rightarrow u, u, e \rightarrow n_1$

TRANSITION ACROSS LABOR STATES

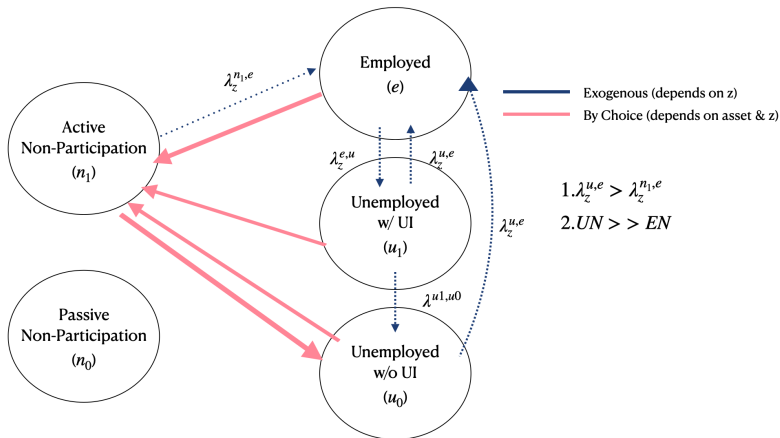


TRANSITION ACROSS LABOR STATES

$\lambda_z^{s,s'}$: Rate from state s to s'

Non-LF Participation

LF Participation

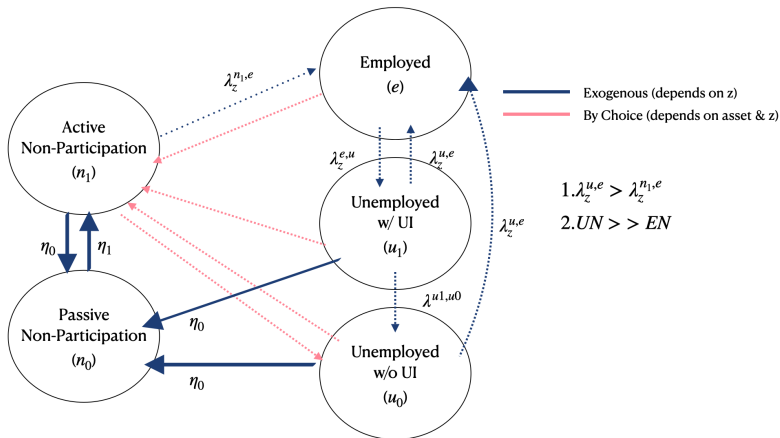


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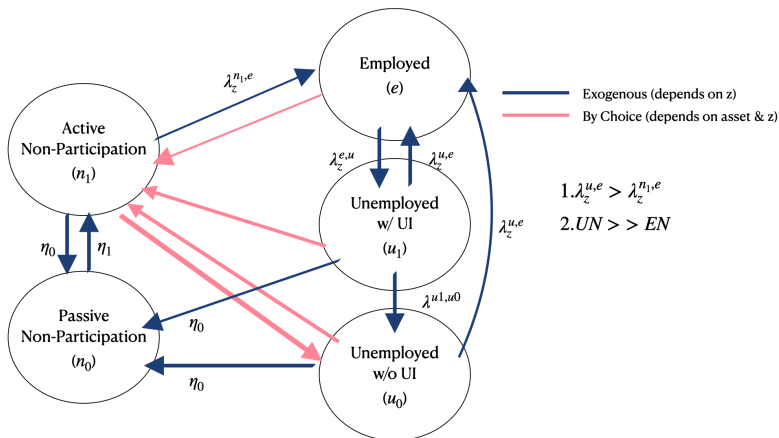
TRANSITION ACROSS LABOR STATES

Policy

$\lambda_z^{s,s'}$: Rate from state s to s'

Non-LF Participation

LF Participation



- State-dependent skill dynamics:

$$d \log Z_t = \left\{ -\theta \log Z_t + \mathbb{I}_{\{s_t=e\}} \delta_Z^+ - \mathbb{I}_{\{s_t \neq e\}} \delta_Z^- \right\} dt + \sigma_Z dW_t$$

- Mean reversion of the skill
- If employed, skill increases
- If unemployed, skills depreciate, leading to lower job finding rate and higher separation rate (Channel 2: Human capital dynamics)

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- State-dependent skill dynamics: Channel 2

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INDIVIDUAL PROBLEM

- Period utility: Bellman

$$u^s(c, h) = \log c - \psi \frac{h^{1+\frac{1}{\sigma}}}{1+\frac{1}{\sigma}} - \kappa^s, \quad s \in \{e, u_1, u_0, n_0, n_1\}$$

Assumption: $\kappa^e > \kappa^u > \kappa^n \geq 0$

- Budget constraint:

$$\begin{aligned} c_t + \dot{a}_t &= r_t a_t + \phi_t + (1 - t_t) w_t z_t h_t, & \text{if } s = e \\ c_t + \dot{a}_t &= r_t a_t + \phi_t + (1 - t_t) b(z_t), & \text{if } s = u_1 \\ c_t + \dot{a}_t &= r_t a_t + \phi_t, & \text{if } s \in \{u_0, n_0, n_1\} \end{aligned}$$

- Borrowing constraint: $a_t \geq 0$
- Choice of individual with state (s, a, z) :
 - Consumption/Saving (optimal control) Participation (optimal stopping)

PARTICIPATION DECISION OVER THE STATE SPACE

- Optimal policy splits state space into participation and non-participation regions [Flow Diagram](#)
- Participation more likely if currently productive (substitution effect) or poor (wealth effect)



FIGURE: Employment to Non-Participation Decision

- A competitive final-good producer aggregates intermediate inputs
- Monopolistic producers with flexible price and linear technology
 - Production requires hiring labor on a continuum of tasks indexed by k
 - A continuum of labor union for each k (sticky wage á la Erceg et al. 2000)

$$\max_{p_t, \{\ell_{kt}\}} \left(\frac{p_t}{P_t} \right) y_t - \int_0^1 w_{kt} \ell_{kt} dk - \chi$$

$$\text{s.t. } y_t = \alpha \ell_t$$

$$\ell_t = \left[\int_0^1 \ell_{kt}^{\frac{\varepsilon-1}{\varepsilon}} dk \right]^{\frac{\varepsilon}{\varepsilon-1}}$$

$$y_t = \left(\frac{p_t}{P_t} \right)^{-\nu} Y_t$$

Solving it yields:

$$\ell_{kt} = \left(\frac{w_{kt}}{w_t} \right)^{-\varepsilon} \ell_t$$

$$\frac{p_t}{P_t} = \frac{\nu}{\nu - 1} \left(\frac{w_t}{\alpha} \right)$$

In symmetric equilibrium,

$$\ell_t = \left(\frac{1}{\alpha} \right) Y_t, \quad w_t = \frac{\alpha(\nu - 1)}{\nu}$$

- CRS production and flexible price implies constant real wage
- which in turn implies $\pi_t^w = \pi_t$

Wedge

- For wage setting, follow Erceg et al. (2000) [Details](#)
 - Every worker i at date t supplies hours on each task k
 - Union can set the wage inflation subject to the quadratic adjustment cost à la Rotemberg(1982) and each employed worker supplies same hours
 - The union maximizes the welfare of a 'stand-in' representative worker
- The wage Phillips curve takes the form

$$\rho(\pi_t - \pi^*) = \frac{\varepsilon}{\Theta} \left[\psi H_t^{1+\sigma} - \left(\frac{\varepsilon - 1}{\varepsilon} \right) (C_t^e)^{-\gamma} (1 - t_t) w_t H_t z_t^e \right] + \dot{\pi}_t$$

- Fiscal & Monetary Policy Fiscal Policy

- Government issues debt, taxes, lump-sum transfers and pays UI benefits
- Central Bank sets the nominal rate based on a policy rule

$$i_t = \max(0, i^* + 1.25 (\pi_t - \pi^{**}) + 0.05 \log \left(\frac{Y_t}{Y^*} \right))$$

EquilibriumAsset

REMAINING INGREDIENTS

- Sources of Fluctuations
 - Demand shock (wedge to the Euler equation)
 - Supply shock (wedge to the wage Phillips Curve)
- Labor Frictions Fit
 - Estimate the labor market transition to match CPS data

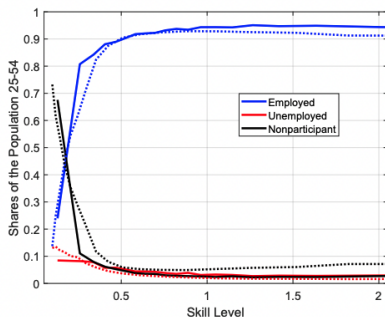
$$\lambda^{ss'}(z) = \lambda_0^{ss'} + \lambda_1^{ss'} \exp\left(\lambda_2^{ss'} z\right)$$

i.e., estimating

$$\left(\lambda_0^{ss'}, \lambda_1^{ss'}, \lambda_2^{ss'}\right)$$

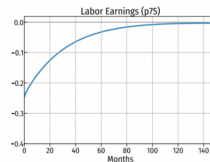
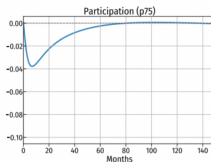
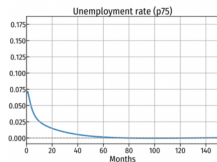
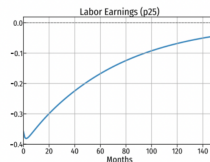
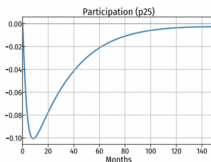
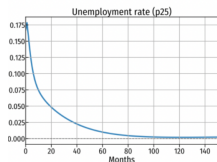
MODEL FIT

	Data	Model
EU	0.017	0.017
EN	0.011	0.011
UE	0.242	0.304
UN	0.189	0.202
NE	0.065	0.043
NU	0.064	0.077



- The paper matches both average flows and stocks by skill level
- Worker flows by skills: CPS + ASEC Data Calibration Skill
- $U \rightarrow N \gg E \rightarrow N$: Channel 3 (Attachment / Participation Cycle)

THREE CHANNELS AT WORK: IRF TO DEMAND SHOCK



- Fluctuations at P25 larger and more persistent than P75
- Recession hurts more for low-skilled workers

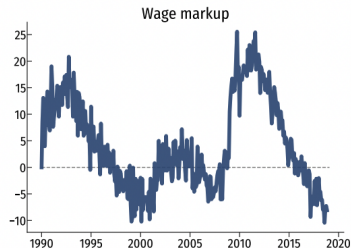
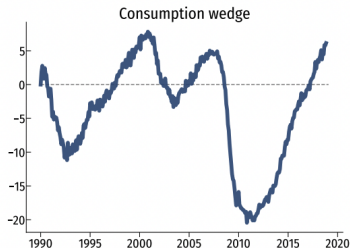
Inclusive Monetary Policies: Counterfactual Analysis

BASLINE MODEL SIMULATION (1990-2019)

- Assume that Fed follows a standard Inflation Target (IT):

$$i_t = \max(0, i^* + 1.25 (\pi_t - \pi^{**}) + 0.05 \log \left(\frac{Y_t}{Y^*} \right))$$

- Estimate the sequence of demand and supply shocks that replicate US employment and inflation 1990–2019 [Solution](#)



COUNTERFACTUAL MONETARY POLICIES

- Counterfactual experiment: How would the US unemployment and inflation dynamics would differ had the Fed adopted more inclusive rule?
- Use the filtered shocks (under IT) to simulate economy under different rules

Baseline Inflation Target	$i_t = i^* + 1.25(\pi_t - \pi^{**}) + 0.05 \log \left(\frac{Y_t}{Y^*} \right)$
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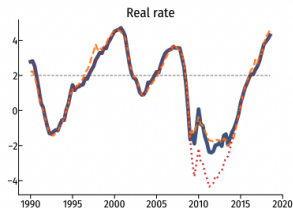
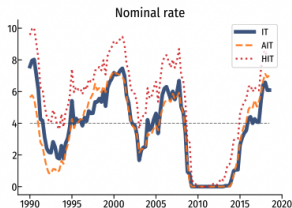
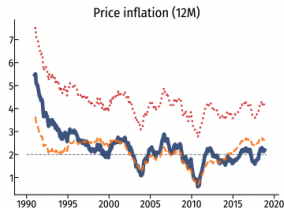
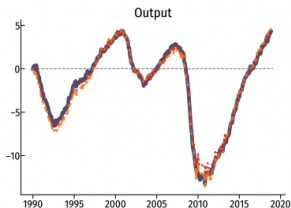
Higher Inflation Target	$i_t = i^{**} + 1.25(\pi_t - \pi^{**}) + 0.05 \log \left(\frac{Y_t}{Y^*} \right)$
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Average Inflation Target	$i_t = i^* + 1.25(\pi_t - \pi^*) + 0.05 \log \left(\frac{Y_t}{Y^*} \right) + 2.00\Gamma_t^\pi$ $\Gamma_t^\pi = (1 - \frac{1}{48})\pi_t + \frac{1}{48}\Gamma_{t-1}^\pi$
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Shortfall Rule	$i_t = i^* + 1.25(\pi_t - \pi^*) + 0.0 \log \left(\frac{Y_t}{Y^*} \right)^+ + 0.05 \log \left(\frac{Y_t}{Y^*} \right)^-$
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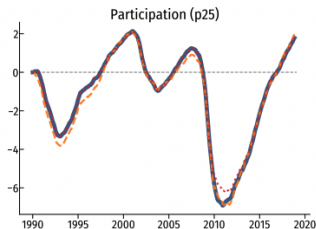
Dovish Rule	$i_t = i^* + 1.25(\pi_t - \pi^*) + 0.0 \log \left(\frac{Y_t}{Y^*} \right)^+ + 0.1 \log \left(\frac{Y_t}{Y^*} \right)^-$
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AIT RULE IS NOT INCLUSIVE



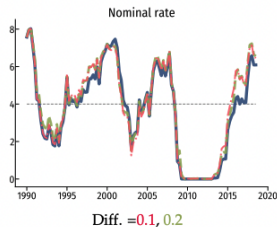
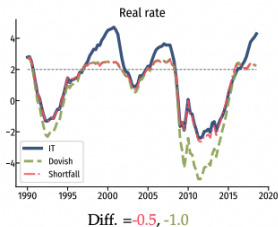
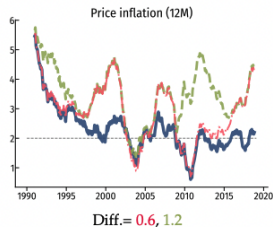
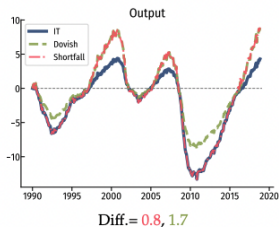
- No significant difference in the dynamics

AIT RULE IS NOT INCLUSIVE



- IT, AIT, and higher trend inflation do not look like inclusive MP

AGGREGATE IMPLICATIONS OF INCLUSIVE RULES

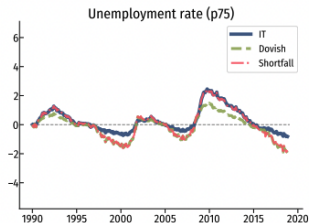


- On average, higher inflation and output, lower real rate

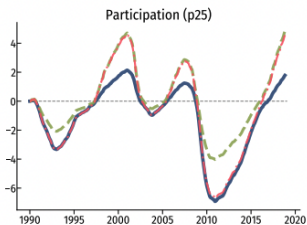
DISTRIBUTIONAL IMPLICATIONS OF INCLUSIVE RULES



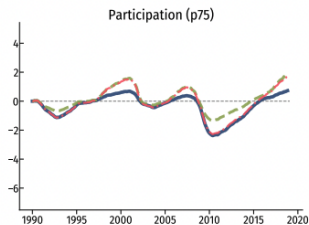
Diff. = -0.6, -1.1



Diff. = -0.2, -0.4

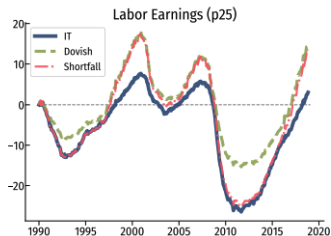


Diff. = 0.6, 1.2

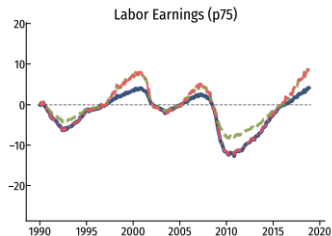


Diff. = 0.2, 0.4

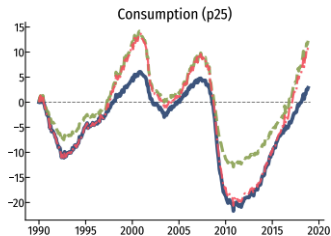
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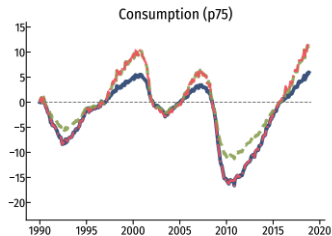
Diff. = 2.7, 5.5



Diff. = 0.8, 1.7



Diff. = 2.1, 4.3



Diff. = 1.00, 2.0

LIMITS: WAGE INDEXATION

- We assumed indexation to the trend inflation:

$$\rho(\pi_t - \pi^*) = \frac{\varepsilon}{\Theta} \left[\psi H_t^{1+\sigma} - \left(\frac{\varepsilon - 1}{\varepsilon} \right) (C_t^e)^{-\gamma} (1 - t_t) w_t H_t z_t^e \right] + \dot{\pi}_t$$

- Wage setters might react to the persistent inflation
 - How about indexation to past inflation?
 - Inflation becomes more persistent, adding up upward pressure to real rates, offsetting the accommodative power

	Index to Trend	Index to Past Inflation
Δ real rate	-0.810	-0.112
Δ output	1.466	0.130
Δ unemployment	-0.555	-0.046
Δ participation	0.549	0.053
Δ unemployment at P25	-0.968	-0.079
Δ earnings at P25	4.675	0.402
Δ consumption at P25	3.648	0.316

CONCLUSION AND FURTHER WORKS

- This paper formalized Okun's hypothesis by adding three channels to the HANK model
- AIT seems not an inclusive rule
 - Asymmetric Taylor rule can run high-pressure economy
 - which improves the labor market prospects of low-skill workers at the cost of high inflation.
- The working paper is not finished. Going forward...
 - The authors plan to decompose the three channels and quantify the contribution
 - Another section on who bears the cost of inflation. Welfare analysis of inclusion-inflation tradeoff

- An interesting paper that formalizes Okun's hypothesis in a modern economic framework
 - An interesting laboratory to examine the distributional consequences of running economy hot
- And I have the following questions:
 - Countries that have run higher inflation do not have lower unemployment
 - Long-run inflation expectation is assumed to be fixed at trend inflation
 - Subject to Lucas Critique and also against international evidence
 - Shocks path obtained under the standard IT rule. But is this really so?
 - All these channels exhaustive and to what extent can we trust the quantitative predictions?

Thank you!

Appendix

BELLMANS OF NON-PARTICIPANTS

- Problem of passive non-participant (n_0) Bellman

$$v_0^{n_0}(a_0, z_0) = \max_{\substack{\{c_t\}_{t \geq 0} \\ \text{s.t.}}} \mathbb{E}_0 \int_0^{\tau^{n_1}} e^{-\rho t} u^n(c_t, h_t) dt + e^{-\rho \tau^{n_1}} v_{\tau^{n_1}}^{n_1}(a_{\tau^{n_1}}, z_{\tau^{n_1}})$$

$$c_t + \dot{a}_t = r_t a_t + \phi_t$$

$$a_t \geq -\bar{a}$$

- Problem of active non-participant (n_1)

$$v_0^{n_1}(a_0, z_0) = \max_{\{c_t\}_{t \geq 0}^{\tau^*}} \mathbb{E}_0 \left[\int_0^{\tau^{\min}} e^{-\rho t} u^n(c_t, h_t) dt \right.$$

$$+ \mathbb{I}_{\{\tau^{\min} = \tau^e\}} e^{-\rho \tau^e} \max \{ v_{\tau^e}^e(a_{\tau^e}, z_{\tau^e}), v_{\tau^{n_1}}^{n_1}(a_{\tau^{n_1}}, z_{\tau^{n_1}}) \}$$

$$\left. + \mathbb{I}_{\{\tau^{\min} = \tau^*\}} e^{-\rho \tau^*} (v_{\tau^*}^{u_0}(a_{\tau^*}, z_{\tau^*}) - \xi) + \mathbb{I}_{\{\tau^{\min} = \tau^{n_0}\}} e^{-\rho \tau^{n_0}} v_{\tau^{n_0}}^{n_0}(a_{\tau^{n_0}}, z_{\tau^{n_0}}) \right]$$

- Problem of unemployed household without UI

$$\begin{aligned}
 v_0^{u_0}(a_0, z_0) = \max_{\{c_t\}_{t \geq 0}} \mathbb{E}_0 \left[\int_0^{\tau^{\min}} e^{-\rho t} u^u(c_t, h_t) dt \right. \\
 \left. + \mathbb{I}_{\{\tau^{\min} = \tau^e\}} e^{-\rho \tau^e} \max \{ v_{\tau^e}^e(a_{\tau^e}, z_{\tau^e}), v_0^{u_0}(a_{\tau^{u_0}}, z_{\tau^{u_0}}) \} \right. \\
 \left. + \mathbb{I}_{\{\tau^{\min} = \tau^*\}} e^{-\rho \tau^*} v_{\tau^*}^{n_1}(a_{\tau^*}, z_{\tau^*}) + \mathbb{I}_{\{\tau^{\min} = \tau^{n_0}\}} e^{-\rho \tau^{n_0}} v_{\tau^{n_0}}^{n_0}(a_{\tau^{n_0}}, z_{\tau^{n_0}}) \right]
 \end{aligned}$$

s.t.

$$c_t + \dot{a}_t = r_t a_t + \phi_t$$

$$a_t \geq -\bar{a}$$

- Problem of unemployed household with UI Bellman

$$\begin{aligned}
 v_0^{u_1}(a_0, z_0) = \max_{\{c_t\}_{t \geq 0}} & \mathbb{E}_0 \left[\int_0^{\tau^{\min}} e^{-\rho t} u^u(c_t, h_t) dt + \mathbb{I}_{\{\tau^{\min} = \tau^e\}} e^{-\rho \tau^e} \max \{ v_{\tau^e}^e(a_{\tau^e}, z_{\tau^e}) \right. \\
 & + \mathbb{I}_{\{\tau^{\min} = \tau^*\}} e^{-\rho \tau^*} v_{\tau^*}^{n_1}(a_{\tau^*}, z_{\tau^*}) + \mathbb{I}_{\{\tau^{\min} = \tau^{u_0}\}} e^{-\rho \tau^{u_1}} v_{\tau^{u_0}}^{u_0}(a_{\tau^{u_0}}, z_{\tau^{u_0}}) \\
 & \left. + \mathbb{I}_{\{\tau^{\min} = \tau^{n_0}\}} e^{-\rho \tau^{n_0}} v_{\tau^{n_0}}^{n_0}(a_{\tau^{n_0}}, z_{\tau^{n_0}}) \right]
 \end{aligned}$$

s.t.

$$c_t + \dot{a}_t = r_t a_t + (1 - t_t) b_t(z_t) + \phi_t$$

$$a_t \geq -\bar{a}$$

- Problems of the employed household Bellman

$$v_0^e(a, z) = \max_{\{c_t\}_{t \geq 0}} \mathbb{E}_0 \left[\int_0^{\tau^{\min}} e^{-\rho t} u^e(c_t, h_t) dt + \mathbb{I}_{\{\tau^{\min} = \tau^u\}} e^{-\rho \tau^u} v_{\tau^u}^{u_1}(a_{\tau^u}, z_{\tau^u}) \right. \\ \left. + \mathbb{I}_{\{\tau^{\min} = \tau^*\}} e^{-\rho \tau^*} v_{\tau^*}^{n_1}(a_{\tau^*}, z_{\tau^*}) \right]$$

s.t.

$$c_t + \dot{a}_t = r_t a_t + (1 - t_t) w_t z_t h_t + \phi_t$$

$$a_t \geq -\bar{a}$$

- Each of the five household problems can be cast recursively as HJB

Simulation

- Use McKay and Wieland (ECMA 2021)'s method to recover shocks only using impulse response
 - Use the Jacobian sequence-space approach of Auclert et al. (2021)
 - Use anticipated monetary shock a la Holden(2016) to include OBC
 - Using recursive filtering method

APPENDIX: WAGE UNION

Labor Union

- Union maximizes the following problem:

$$\max_{\{\omega_{kt}\}_{t \geq 0}} \int_0^{\infty} e^{-\rho t} \left[u^e(C_t^e, H_t) - \frac{\Theta}{2} \left(\frac{\dot{\omega}_{kt}}{\omega_{kt}} - \pi^* \right)^2 \right] dt$$

$$C_t^e \equiv \int_{s_{it}=e} c_{it} di, \quad H_t \equiv \int_{s_{it}=e} h_{it} di, \quad h_{it} \equiv \int_0^1 h_{kt} dk$$

$$c_{it} + \dot{a}_{it} = r_t a_{it} + (1 - t_t) \frac{1}{P_t} z_{it} \int_0^1 \omega_{kt} h_{kt} dk + \phi_t$$

$$h_{kt} \int_{s_{it}=e} z_{it} di = \left(\frac{\omega_{kt}}{\omega_t} \right)^{-\varepsilon} \ell_t$$

APPENDIX: WAGE UNION

- Hour per worker h_{kt} is determined by $h_{kt} \int_{s_{it}=e} z_{it} di = \ell_{kt}$
- If we have the set of working individuals at t , residually derive the hours of work h_{kt} that each employed individual supply.

Model

- G_t : government purchase, ϕ_t : lump sum transfer, b_t : UI benefits, τ_t : labor income tax, $B_t^g > 0$ government debt Model
- The government faces the following intertemporal budget constraint:

$$G_t + \phi_t + (1 - \tau_t) \int_{s_{it}=u^1} b(z_{it}) di + r_t^b B_t^g = \tau_t w_t h_t \int_{s_{it}=e} z_{it} di + \dot{B}_t^g$$

- The fiscal policy rule is passive:

$$\tau_t = \tau^* + \beta_B (B_t^g - B^*), \quad \beta_B > 1$$

MUTUAL FUNDS

- Household wealth equals the shares of the mutual fund
- The mutual fund chooses optimal portfolio between bonds and equity (of int. good firms)
- Establishes a no-arbitrage condition between government bonds and firm equity except when there is shock $r_t = \frac{\Pi_t + \dot{q}_t}{q_t} = r_t^b$
- Asset market clearing:

$$\underbrace{\sum_{s \in \{e, u, n\}} \int a_t d\mu_t^s}_{\text{sum of asset holdings}} = \underbrace{A_t}_{\text{the value of mutual fund}} = \underbrace{q_t + B_t^g}_{\text{equity price plus gov. debt}}$$

Model

CALIBRATION

Model

Parameter		Value	Target
<i>Preferences</i>			
Discount rate	ρ	0.014	Mean wealth ($0.56 \times$ annual earnings)
Relative risk aversion	γ	1	External
Labor supply elasticity	σ	1	External
Weight on hours worked	ψ	0.82	Normalize hours $h = 1$
Utility cost of working	κ^e	0.87	NE flow (0.011)
Utility cost of job searching	κ^u	0.10	Mean time spent in job search
<i>Productivity Dynamics</i>			
Diffusion of log-productivity	σ_z	0.05	Literature
Autocorrelation of log-productivity	$\exp(-\theta)$	0.25	Literature
Skill drift while employed	δ^+	0.4	Normalize $\left(\frac{1}{E_t} \int_{t_H=t}^{\infty} z_{it} di\right)$
Skill drift while non-employed	δ^-	5.1	Earnings losses upon displacement
<i>Labor market transitions</i>			
Job destruction rate	$\lambda_0^{eu} + \lambda_1^{eu} \exp(\lambda_2^{eu} z)$		eu flow (0.017)
Job finding rate out of U	$\lambda_0^{ue} + \lambda_1^{ue} \exp(\lambda_2^{ue} z)$		ue flow (0.242)
Job finding rate out of N	$\lambda_0^{ne} + \lambda_1^{ne} \exp(\lambda_2^{ne} z)$		ne flow (0.065)
Transition rate to passive N	η_0		un flow (0.189)
Transition rate to active N	η_1		nu flow (0.064)
<i>Technology</i>			
Productivity	α	1.38	Normalize real wages $w = 1$
Elasticity of substitution	ν	10	Literature
Elasticity of substitution	ε	10	Literature
Fixed cost	χ	0.088	Interest rate of 2%
Nominal wage adjustment cost	Θ	4500	Slope of Phillips curve (Del Negro et al., 2020)

Motivation

- New Economic Challenges and the Fed's Monetary Policy Review
 - Our revised statement emphasizes that maximum employment is a broad-based and inclusive goal. This change reflects our appreciation for the benefits of a strong labor market, particularly for many in low- and moderate-income communities. In addition, our policy decision will be informed by our "assessments of the shortfalls of employment from its maximum level" rather than by "deviations from its maximum level" as in our previous statement.

An equilibrium is:

- household consumption decisions $\{c_t^s(a, z)\}_{t \geq 0}$ for $s \in \{e, u_0, u_1, n_0, n_1\}$
- participation and job offer acceptance decisions $\{p_t^s(a, z), f_t^s(a, z)\}_{t \geq 0}$ for $s \in \{e, u_0, u_1, n_1\}$
- unions' nominal wage setting $\{\omega_{kt}\}_{t \geq 0}$ for all labor types k
- intermediate producers' hiring decisions $\{\ell_{kt}\}_{t \geq 0}$ for all k
- mutual fund allocations between equity and government bonds $\{X_t^m, B_t^m\}_{t \geq 0}$
- real rates of return on the mutual fund and on government bonds $\{r_t, r_t^b\}_{t \geq 0}$, firms' share price $\{q_t\}_{t \geq 0}$
- fiscal variables (taxes, transfers, UI benefits, expenditures and debt) $\{t_t, \phi_t, b_t(z), G_t, B_t^g\}_{t \geq 0}$

- nominal interest rates $\{i_t\}_{t \geq 0}$, aggregate output, consumption, profits, contractual hours worked, and inflation $\{Y_t, C_t, \Pi_t, h_t, \pi_t\}_{t \geq 0}$
- measures of households $\{\mu_t^s(a, z)\}_{t \geq 0}$ for $s \in \{e, u_0, u_1, n_0, n_1\}$

such that at every t : (i) households maximize their utility; (ii) final good and intermediate good producers maximize their profits; (iii) inflation satisfies the Phillips curve; (iv) the government budget constraint holds; (v) all good and asset markets clear.

Model

- Data Sources

- Basic Monthly CPS: Provides employment status (1989–2019)
- ASEC: Supplemented to provide individual labor earnings data
- Sample: Male individuals between 25 and 55 years of age

- Variable Definition

- Skill Level: Proxy for skill level is weekly real earnings
- Transition Rates: Separation and job finding rates based on month-to-month changes in employment status by each skill level

- Estimation Methodology

- Minimum quadratic distance estimator to minimize the difference between empirical data and model specification:

$$\lambda^{ss'}(z) = \lambda_0^{ss'} + \lambda_1^{ss'} \exp\left(\lambda_2^{ss'} z\right)$$

- Estimates transition parameters for different phases of the business cycle (boom, expansion, recession, crisis) Model

LABOR FLOW FITTING

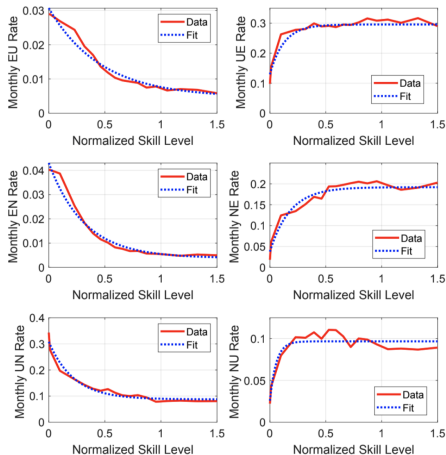
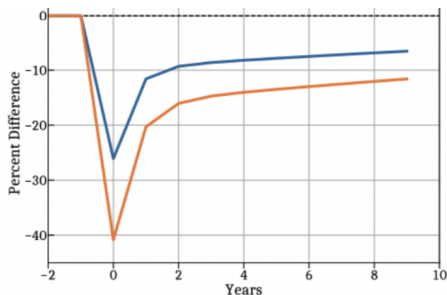
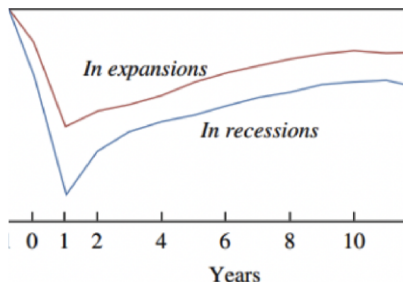


FIGURE: Empirical transition rates vs. Model fit

Model1

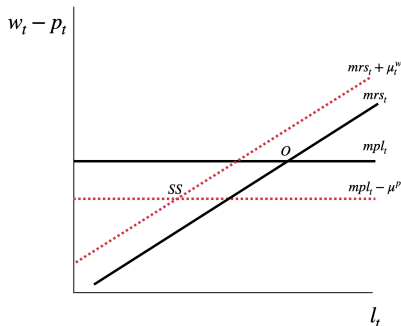
Model2

SKILL DYNAMICS CALIBRATION



- Recall $d \log z_t = \left\{ -\theta \log z_t + \mathbb{I}_{\{s_t=e\}} \delta_z^+ - \mathbb{I}_{\{s_t \neq e\}} \delta_z^- \right\} dt + \sigma_z dW_t$
- Upon displacement, losses are large, persistent and countercyclical

Model



- Flat MPL due to CRS technology Model
- Wedge to labor demand constant, wedge to labor supply variable
 - Both are wedges so similar effect on real economy.
 - Variability of price and wage inflation is different (Smets and Wouters, AER 2007)